BUYING THE RIGHT LED TUBELIGHT

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Light emitting diodes (LEDs) are on the verge of becoming the standard lighting technology today. Households and offices alike have recognised their ability to reduce electricity bills and deliver collateral benefits such as better colour options and higher lifespans over fluorescent and incandescent lights. Among LED lightings, tubelights is one of the most popular categories. As the market is filling up with numerous choices for LED tubelights, it is important to know the parameters that can help you pick the right one for you.

Important characteristics of LED tubelights

In this section we discuss the various features you need to look out for in an LED tubelight. These features will help you determine the brightness and longevity of the tubelight along with the power it will consume.

Lumens for brightness. Lumens is a unit that measures the intensity of brightness that a light can generate. Higher the lumen value, brighter the light.

For households, tubelights with 1600-lumen capacity are enough. Whereas, for office premises, tubelights must have at least 2000-lumen brightness.

Wattage and efficacy for power consumption. Wattage signifies the amount of electricity a tubelight will consume to deliver brightness (in lumens). Also, higher wattage means higher electricity bills! Usually, 20W is sufficient for the lumen capacity mentioned earlier and, hence, is preferable to be restricted to that value.

Luminous efficacy is determined by lumen-to-wattage ratio of the tubelight. This is a direct indication of the brightest tubelight consuming the lowest energy.

Domestic tubelights should have an efficacy higher than 85 lumens per watt, while those for the office must have an efficacy higher than 100 lumens per watt. Higher efficiency indicates a more economical tubelight in the long run.

Colour temperature for proper ambience. LED tubelights come in different colours, and are measured by their colour temperature—sometimes called colour correlation temperature (CCT)—in Kelvin (K).

Warm yellow light, meant for a soothing and relaxing ambience, can be identified by a colour temperature of around 3000K. On the other hand, bright cool blue or white light can be identified by a colour temperature of over 5000K. Common white light is identifiable at around 6500K.

However, it should be noted that a higher colour temperature can often be accompanied by a strong glare. To avoid this, check for milky diffused property in tubelights, where emitted light is dampened slightly by the diffusor. On the other hand, in case of a clear diffusor, complete intensity of the light is emitted.